Path integral

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未分类

Weinberg: In canonical quantization: Yukawa $\partial_\mu\bar{\psi}\gamma^\mu\gamma^5\psi$ (vector field) → straight forward but rather awkward.

$\mathcal{H}_I$ always contain an additional non-covariant term ⇒ Cancel non-covariant terms in the propagator

QED: Can bear the awkwardness

Non-Abelian gauge theories: unbearable.

Path-Integral: directly from Lagrangian. to the Feynman rules
Lorentz-covariant form.

Feynman: inspired by Dirac.

Schwinger and Tomonaga: Operator methods Most choose in 1950s.

Revived in the late 1960s (Faddeev, Popov, and De Witt apply it to non-Abelian gauge theories and general relativity).

1971 t'Hooft derive Feyman rules for spontaneously broken gauge theories.

Why the S-matrix calculated in this way is unitory?

directly from Path integral may be wrong eg. non-linear σ-model. $\mathcal{L} = -\frac{1}{2}f_{\pi\pi}(\phi)\,\partial_\mu\phi^\alpha\partial^\mu\phi^\beta$.

Weinberg: Derive the path-integral formlism from the canonical formlism!

General Path-Integral Formula.

$[Q_a, P_b] = i\delta_{ab}$ $\quad [Q_a, Q_b] = [P_a, P_b] = 0$.

In field theory: $Q_{x,m} \equiv Q_m(\vec{x})$ $\quad \delta_{x,m;\vec{y},n} \equiv \delta^3(\vec{x}-\vec{y})\,\delta_{mn}$. Schrödinger-picture. operators. $t=0$.

$P_{x,m} \equiv P_m(\vec{x})$

Q_a a: x, m $\quad [Q_a, Q_b] = 0$ ⇒

A simultaneous eigenstate $|q\rangle$ $\quad Q_a|q\rangle = q_a|q\rangle$

$\langle q'|q\rangle = \prod_a \delta(q_a' - q_a) \equiv \delta(q'-q)$

$1 = \int \prod_a dq_a\,|q\rangle\langle q|$.

$P_a|p\rangle = p_a|p\rangle$ $\quad \langle p'|p\rangle = \prod_a \delta(p_a' - p_a) \equiv \delta(p'-p)$ $\quad 1 = \int \prod_a dp_a\,|p\rangle\langle p|$

$\langle q|[Q_a, P_b]|q'\rangle = q_a\langle q|P_b|q'\rangle - q_a'\langle q|P_b|q'\rangle$

$i\delta_{ab}\langle q|q'\rangle = (q_a - q_a')(\langle q|P_b|q'\rangle)$ ……

$\delta_{ab}\langle q|q'\rangle = -i(q_a - q_a')\langle q|P_b|q'\rangle$

$\exists a=b$ 时 $\quad \langle q|q'\rangle = -i(q_a - q_a')\langle q|P_a|q'\rangle$

$\delta(q-q') = -i(q_a - q_a')\langle q|P_a|q'\rangle$

$= -(q_a - q_a')\frac{\partial}{\partial q_a}\delta(q_a - q_a')\prod_{b\neq a}\delta(q_b - q_b')$

$\Rightarrow \langle q|P_a|q'\rangle = -i\frac{\partial}{\partial q_a}\langle q|q'\rangle$ ..

$\Rightarrow \langle q|p\rangle = \prod_a \frac{1}{\sqrt{2\pi}}\exp(iq_a p_a)$

Heisenberg picture:

$Q_a(t) \equiv \exp(iHt)\,Q_a\exp(-iHt)$ $\quad Q_a(t)|q;t\rangle = q_a|q;t\rangle$

$P_a(t) \equiv \exp(iHt)\,P_a\exp(-iHt)$ $\quad P_a(t)|p;t\rangle = p_a|p;t\rangle$

$\Rightarrow$ $\begin{cases} |q;t\rangle = \exp(iHt)|q\rangle \\ |p;t\rangle = \exp(iHt)|p\rangle \end{cases}$

$|q;t\rangle$: the eigenstate of $\hat{Q}_a(t)$ with eigenvalue q_a.

$\begin{cases} |q;t\rangle = \exp(iHt)|q\rangle \\ |p;t\rangle = \exp(iHt)|p\rangle \end{cases}$

$\langle q';t|q;t\rangle = \delta(q'-q)$ $\quad$ $\langle p';t|p;t\rangle = \delta(p'-p)$

$\int \prod_a dq_a\, |q;t\rangle\langle q;t| = 1$ $\quad$ $\int \prod_a dp_a\, |p;t\rangle\langle p;t| = 1$

$\langle q;t|p;t\rangle = \prod_a \frac{1}{\sqrt{2\pi}} \exp(iq_a p_a)$

$\langle q';z+dz|q;z\rangle = \langle q';z|\exp(-iH dz)|q;z\rangle$ $\quad$ $H \equiv H(Q,P) = e^{iHt} H(Q,P) e^{-iHt} = H(Q(t),P(t))$

Q in P's Left.

$1 - i\hat{H}(Q(z),P(z))$

$\langle q';z+dz|q;z\rangle = \int \prod_a dp_a \langle q';z|\exp(-iH(Q(z),P(z))dz)|p;z\rangle \times \langle p;z|q;z\rangle$

$= \int \prod_a \frac{dp_a}{2\pi} \exp\left[-iH(q',p)dz + i\sum_a (q'_a - q_a)p_a\right]$ $\quad$ for $z \to 0$.

The more rigorous version can be seen in. Naxahara_Geometry, topology, and physics.

$\langle q';t'|q;t\rangle$ $\quad$ $t, z_1, z_2, \cdots z_N, t'$

$z_{k+1} - z_k = dz = (t'-t)/(N+1)$

$\langle q';t'|q;t\rangle = \int dq_1 \cdots dq_N \langle q';t'|q_N;z_N\rangle\langle q_N;z_N|q_{N-1};z_{N-1}\rangle \cdots \langle q_1;z_1|q;t\rangle$

$= \int \prod_{k=1}^{N} \prod_a dq_{k,a} \prod_{k=0}^{N} \prod_a \frac{dp_{k,a}}{2\pi} \exp\left[i \sum_{k=1}^{N+1} \left\{\sum_a (q_{k,a} - q_{k-1,a})p_{k-1,a} - H(q_k, p_{k-1})dz\right\}\right]$.

$q_0 \equiv q$ $\quad$ $q_{N+1} \equiv q'$ $\quad$ $q_a(z_k) \equiv q_{k,a}$ $\quad$ $p_a(z_k) \equiv p_{k,a}$.

$\sum_{k=1}^{N+1} \left\{\sum_a (q_{k,a} - q_{k-1,a})p_{k-1,a} - H(q_k, p_{k-1})dz\right\} = \sum_{k=1}^{N+1} \left\{\sum_a \dot{q}_a(z_k) p_a(z_k) - H(q(z_k), p(z_k))\right\} dz + O(dz^2)$

$\to \int_t^{t'} \left\{\sum_a \dot{q}_a(z) p_a(z) - H(q(z), p(z))\right\} dz$.

$\int \prod_{z,a} dq_a(z) \prod_{z,b} \frac{dp_b(z)}{2\pi} \cdots \equiv \lim_{dz \to 0} \int \prod_{k,a} dq_{k,a} \prod_{k,b} \frac{dp_{k,b}}{2\pi} \cdots$

$\langle q';t'|q;t\rangle = \int_{\substack{q_a(t)=q_a \\ q_a(t')=q'_a}} \prod_{z,a} dq_a(z) \prod_{z,b} \frac{dp_b(z)}{2\pi} \times \exp\left[i\int_t^{t'} dz \left\{\sum_a \dot{q}_a(z) p_a(z) - H(q(z), p(z))\right\}\right]$

$\langle q';z+dz|O(p(z),Q(z))|q;z\rangle = \int \prod_a dp_a \times \langle q';z|\exp(-iH(Q(z),P(z))dz)|p;z\rangle\langle p;z|O(P(z),Q(z))|q;z\rangle$

$= \int \prod_a \frac{dp_a}{2\pi} \exp\left[-iH(q',p)dz + i\sum_a (q'_a - q_a)p_a\right] O(p,q)$

$\langle q';t'|O_A(P(t_A),Q(t_A))O_B(P(t_B),Q(t_B))\cdots|q,t\rangle$

$= \int_{\substack{q_a(t)=q_a \\ q_a(t')=q'_a}} \prod_{z,a} dq_a(z) \prod_{z,b} \frac{dp_b(z)}{2\pi} O_A(p(t_A),q(t_A))O_B(p(t_B),q(t_B))\cdots \times \exp\left[i\int_t^{t'} dz \left\{\sum_a \dot{q}_a(z)p_a(z) - H(q(z),p(z))\right\}\right]$

$t' > t_A > t_B > \cdots > t$.

$\langle q',t'|T\{O_A(P(t_A),Q(t_A)), O_B(P(t_B),Q(t_B)),\cdots\}|q,t\rangle$

$= \int_{\substack{q_a(t)=q_a \\ q_a(t')=q'_a}} \prod_{z,a} dq_a(z) \prod_{z,b} \frac{dp_b(z)}{2\pi} O_A(p(t_A),q(t_A))O_B(p(t_B),q(t_B))\cdots \times \exp\left[i\int_t^{t'} dz \left\{\sum_a \dot{q}_a(z)p_a(z) - H(q(z),p(z))\right\}\right]$

if $O_A(p(t_A),q(t_A)) = \dot{q}_a(t_A) - \frac{\partial H(q(t_A),p(t_A))}{\partial p_a(t_A)}$ or $\dot{p}_a(t_A) + \frac{\partial H(q(t_A),p(t_A))}{\partial q_a(t_A)}$

$O_A(p(t_A),q(t_A))\exp(iI[q,p]) = -i\frac{\delta}{\delta p_a(t_A)}\exp(iI[q,p])$

$\left(i\frac{\delta}{\delta q_a(t_A)}\exp(iI[q,p])\right)$

$I[q,p] \equiv \int_t^{t'} dz \left\{\sum_a \dot{q}_a(z)p_a(z) - H(q(z),p(z))\right\}$

if $t < t_B < t'$ The path-integral will be divergent.

Hence if $O_A(p(t_A),q(t_A))$ $\leftarrow$ The path integral must vanish.

if t_A approaches t_B or t_C etc The path-integral $\propto \delta(t_A - t_B)$ or $\frac{\partial}{\partial t_A}\delta(t_A - t_B)$

$a : \vec{x}, m$

$\langle q';t'|T\{O_A[P(t_A),Q(t_A)], O_B(P(t_B),Q(t_B)], \cdots\}|q,t\rangle$

$= \int_{q_m(\vec{x},t)=q_m(\vec{x})} \prod_{z,\vec{x},m} dq_m(\vec{x},z) \prod_{z,\vec{x},m} \frac{dp_m(\vec{x},z)}{2\pi} O_A[p(t_A),q(t_A)]O_B[p(t_B),q(t_B)]\cdots$

$q_m(\vec{x},t')=q'_m(\vec{x})$ $\quad \times \exp\left[i\int_t^{t'} dz \int d^3x \sum_m \dot{q}_m(\vec{x},z)\pi_m(\vec{x},t) - H[q(z),p(z)]\right]$

In a measurement. not $|q',t'\rangle$ or $|q,t\rangle$

$t\to-\infty$ $t\to+\infty$ $|\alpha,in\rangle$ $|\beta,out\rangle$

α,β denote sets of particles characterized by the various particles' momenta, spin-z-component (or helicity) and species.

Need 'wave fuctions' $\langle\beta,out|q',t'\rangle$ $\langle q,t|\alpha,in\rangle$ For convenience $\begin{cases} t'\to\infty \\ t\to-\infty. \end{cases}$

$\pi_m(\vec{x},+\infty)=q'_m(\vec{x})$ $\quad \pi_m(\vec{x},-\infty)=\pi_m(\vec{x})$

Integrating over $q'_m(\vec{x})$ and $q_m(\vec{x})$

$\langle\beta,out|T\{\mathcal{O}_A[P(t_A),Q(t_A)],\mathcal{O}_B[P(t_B),Q(t_B)],\cdots\}|\alpha,in\rangle$

$=\int \prod_{\tau,\vec{x},m} d\pi_m(\vec{x},\tau) \prod_{\tau,\vec{x},m} (d p_m(\vec{x},\tau)/2\pi) \times \mathcal{O}_A[P(t_A),q(t_A)]\mathcal{O}_B[P(t_B),q(t_B)]\cdots$

$\times \exp\left[i\int_{-\infty}^{+\infty} dz \left\{\int d^3x \sum_m \dot{q}_m(\vec{x},z)p_m(\vec{x},t) - H[q(z),p(z)]\right\}\right]$

$\times \langle\beta,out|q;+\infty\rangle\langle q;-\infty|\alpha,in\rangle.$

How to calculate $\langle q;-\infty|\alpha,in\rangle$ $\langle q;+\infty|\beta,out\rangle$?

Assume: For $t\to\pm\infty$, No interactions.

$a_{in}(\vec{p},\sigma,n)|VAC,in\rangle=0$

$a_{out}(\vec{p},\sigma,n)|VAC,out\rangle=0.$

a_{in}, a_{out} : coefficients of $\exp(i\vec{p}\cdot\vec{x}-iEt)$ in the plane-wave expansion of the operator $q_m(\vec{x},t)$ at $t\to-\infty$ and $t\to+\infty$

eg. real scalar field.

$\Phi(\vec{x},t) \xrightarrow[t\to\mp\infty]{} (2\pi)^{-3/2}\int d^3p\,(2E)^{-1/2}\left[a^{in}_{out}(\vec{p})e^{ipx}+H.c.\right]$

$\Pi(\vec{x},t) = \dot{\Phi}(\vec{x},t) \xrightarrow{t\to\mp\infty} -i(2\pi)^{-3/2}\int d^3p\,(E/2)^{1/2}\left[a^{in}_{out}(\vec{p})e^{ipx}-H.c.\right]$

$p^0\equiv E\equiv\sqrt{\vec{p}^2+m^2}.$

$a^{in}_{out}(\vec{p}) = \lim_{t\to\mp\infty}\frac{e^{iEt}}{(2\pi)^{3/2}}\int d^3x\, e^{-i\vec{p}\cdot\vec{x}}\left[\sqrt{\frac{E}{2}}\Phi(\vec{x},t)+i\sqrt{\frac{1}{2E}}\Pi(\vec{x},t)\right].$

So: $0=\int d^3x\, e^{-ipx}\left[\underline{\frac{\delta}{\delta\phi(x)}} \mp E(\vec{p})\phi(\vec{x},t)\right]\langle\phi(t\to\mp\infty);\mp\infty|VAC.{}^{in}_{out}\rangle$

has a well-known Gaussian solution:

$\langle\phi(t\to\mp\infty);\mp\infty|VAC,{}^{in}_{out}\rangle = \mathcal{N}\exp\left(-\frac{1}{2}\int d^3x d^3y\, \mathcal{E}(\vec{x},\vec{y})\phi(\vec{x})\phi(\vec{y})\right)$

$0=\int d^3x\, e^{-i\vec{p}\cdot\vec{x}}\left[\int d^3y\, \mathcal{E}(\vec{x},\vec{y})\phi(\vec{y}) - E(\vec{p})\phi(\vec{x})\right]$

$\int d^3x\, e^{-i\vec{p}\cdot\vec{x}}\mathcal{E}(\vec{x},\vec{y}) = E(\vec{p})e^{-i\vec{p}\cdot\vec{y}}$

$\mathcal{E}(\vec{x},\vec{y}) = (2\pi)^{-3}\int d^3p\, e^{i\vec{p}\cdot(\vec{x}-\vec{y})}E(\vec{p})$ $\quad$ 而 $E(\vec{p})\equiv\sqrt{\vec{p}^2+m^2}$

$\Rightarrow \mathcal{E}(\vec{x},\vec{y}) = \frac{m}{2\pi^2 r}\frac{d}{dr}\left(\frac{1}{r}K_1(mr)\right)$ $\quad r=|\vec{x}-\vec{y}|$

$\langle VAC,out|\phi(\vec{x});+\infty\rangle\langle\phi(\vec{y});-\infty|VAC,in\rangle$

$=|\mathcal{N}|^2\exp\left(-\frac{1}{2}\int d^3x d^3y\, \mathcal{E}(\vec{x},\vec{y})\left[\phi(\vec{x},+\infty)\phi(\vec{y},+\infty)+\phi(\vec{x},-\infty)\phi(\vec{y},-\infty)\right)\right.$

$f(+\infty)+f(-\infty) = \lim_{\epsilon\to0+} c\int_{-\infty}^{\infty} dz f(z)e^{-\epsilon|z|}$

$\Rightarrow \langle VAC,out|T\{\mathcal{O}_A[\Pi(t_A),\dot{\Phi}(t_A)],\mathcal{O}_B(\Pi(t_B),\dot{\Phi}(t_B)],\cdots\}|VAC.in\rangle$

$=|\mathcal{N}|^2\int_{t,\vec{x}}\prod d\phi(\vec{x},t)\prod_{\tau,\vec{x}}(d\pi(\vec{x},\tau)/2\pi)\,\mathcal{O}_A[\pi(t_A),\phi(t_A)]$

$\times\mathcal{O}_B[\pi(t_B),\phi(t_B)]\cdots\exp\left[i\int_{-\infty}^{\infty} d\tau\int d^3x\,\dot{\phi}(\vec{x},\tau)\pi(\vec{x},\tau)-H[\phi(\tau),\pi(\tau)]\right.$

$\left.\pm\frac{1}{2}i\epsilon\int d^3x d^3y\,\mathcal{E}(\vec{x},\vec{y})e^{-\epsilon|\tau|}\phi(\vec{x},\tau)\phi(\vec{y},\tau)\right\}]$

$i\epsilon$ terms.

When calculating the matrix elements between multi-particle states, by inserting the appropriate 'wave functionals' in Eq.(9.2.3)', Applying the adjoints of annihilation operators. a, $a^\dagger$

Lagrangian Version of the Path-Integral Formula.

For theories with a Hamiltonian that is quadratic in the momenta

$$H[q,p]=\frac{1}{2}\sum_{nm}\int d^3x d^3y\, A_{\vec{x}n,\vec{y}m}[q]\,p_n(\vec{x})\,p_m(\vec{y})+\sum_n\int d^3x\, B_{\vec{x}n}[q]\,p_n(\vec{x})+C[q].$$

$A_{\vec{x}n,\vec{y}m}$ is real, symmetric, positive, and non-singular.

$$\int dt\Big\{\int d^3x\sum_n p_n(\vec{x},t)\,\dot{q}_n(\vec{x},t)-H[q(t),p(t)]\Big\}$$
$$=-\frac{1}{2}\sum_{mn}\int d^3x d^3y\,dt dt'\; A_{t\vec{x}n,t'\vec{y}m}[q]\;p_n(\vec{x},t)\,p_m(\vec{y},t')-\sum_n\int d^3x\int dt\; B_{t\vec{x}n}[q]\,p_n(\vec{x},t)-B[q].$$

$$A_{t\vec{x}n,t'\vec{y}m}[q]\equiv A_{\vec{x}n,\vec{y}m}[q(t)]\,\delta(t-t')$$

$$B_{t\vec{x}n}[q]\equiv B_{\vec{x}n}[q(t)]-\dot{q}_n(\vec{x},t)$$

$$C[q]\equiv\int dt\, C[q(t)].$$

$$\int_{-\infty}^{\infty}\Big(\prod_s d\xi_s\Big)\exp\Big\{-\frac{1}{2}i\sum_{sr}A_{sr}\xi_s\xi_r-i\sum_s B_s\xi_s-iC\Big\}$$
$$=(\mathrm{Det}[iA/2\pi])^{-1/2}\exp\Big\{-i\frac{1}{2}\sum_{sr}A_{sr}\bar{\xi}_s\bar{\xi}_r-i\sum_s B_s\bar{\xi}_s-iC\Big\}$$

$$\bar{\xi}_s=-\sum_r(A^{-1})_{sr}B_r.$$

if ϑ_A,ϑ_B are independent of P_s

We can evaluate the path integral over the ps

Just setting these variables at the stationary point

$$\frac{\delta}{\delta p_n(\vec{x},t)}\int_{-\infty}^{\infty}dt\Big\{\int d^3x\,\dot{q}_n(\vec{x},t)\,p_n(\vec{x},t)-H[q(t),p(t)]+\text{ie terms}\Big\}$$
$$=\dot{q}_n(\vec{x},t)-\frac{\delta}{\delta p_n(\vec{x},t)}H[q(t),p(t)]$$

$$\dot{q}_n(\vec{x},t)=\Big[\frac{\delta H[q(t),p(t)]}{\delta p_n(\vec{x},t)}\Big]_{p=\bar{p}}\qquad \bar{p}\text{ are the result of the canonical formula.}$$

$$L[q(t),\dot{q}(t)]\equiv\int d^3x\Big(\sum_n\dot{q}_n(\vec{x},t)\,\bar{p}_n(\vec{x},t)-H[q(t),\bar{p}(t)]\Big)$$

$$\to\ \langle\mathrm{VAC,out}|T\{\mathcal{O}_A[q(t_{A1}),\mathcal{O}_B[q(t_{B1}],\cdots]|\mathrm{VAC,in}\rangle$$
$$=|N^2|\int_{\prod_{i,\vec{x}}n}dq_n(\vec{x},t)\,(\mathrm{Det}[2i\pi\vartheta[q]])^{-1/2}\times\mathcal{O}_A[q(t_{A1}]\,\mathcal{O}_B[q(t_{B1})]\cdot$$
$$\times\exp\Big[i\int_{-\infty}^{\infty}dt\{L[q(t),\dot{q}(t)]+\text{ie terms}\}\Big].$$

Gaussian Multiple Integrals

$$J\equiv\int_{-\infty}^{\infty}\prod_r d\xi_r\,\exp\{-Q(\xi)\}.\qquad Q(\xi)=\frac{1}{2}\sum_{rs}K_{rs}\xi_r\xi_s+\sum_r l_r\xi_r+M$$

K_{rs} $\quad r\leftrightarrow s$ symmetric and non-singular

